#13 · 3.4% of marks

Ch 4 — Security Analysis

Theory-heavy. SM TYK theory Qs: **EMH and its misconception · Dow theory · Elliot wave theory · indicators of economic performance.** Numericals: moving averages / EMA, RSI (ICAI paper), dividend-growth value.

① Fundamental analysis (SM 1)

- → *Share price depends on **expected future dividends**; PV of dividends = **intrinsic value**. Price below intrinsic → **buy**; above → **sell***
- → *Constant dividend: $P_0 = D \div k$. Constant growth ($g < k$): **$P_0 = D_1 \div (k - g) = D_0(1 + g) \div (k - g)$** — **Gordon's dividend growth model** (Myron J. Gordon); invalid if $g \geq k$ (infinite price) — not restrictive, long-term g unlikely to exceed k. $g = 0$ → back to $D \div k$*
- → ***Link to P/E (SM 1.1)**: with payout fraction **b**, $D_1 = bE_1$ → **$P_0 = bE_1 \div (k - g)$**; so $P/E = b \div (k - g)$. Price / P/E above fundamental → overpriced → sell; below → buy*
- → *Three concentric circles: **economy** → **industry** → **company** analysis*

> ⚠ **TRAP**
> ***Symbol trap**: in SM Ch 4, **b = payout** ratio. In SM Ch 5 (Gordon earnings model) and Ch 1 (SGR), **b = retention**. Read the definition given in the question.*

② Economic analysis (SM 1.2)

Past / present / expected performance of the economy and sectors; income & consumption trends drive profits, dividends, prices.

Factors (4)

*(a) growth of **national income** — nominal GDP vs **real** growth (b) growth of **industrial sector** (by demand) (c) **inflation** — WPI / CPI; consumer industries fine-tune pricing (d) **monsoon** — forward & backward linkages*

Techniques (3) — TYK Q4

*(a) **Anticipatory surveys** — expert opinion on construction, capex, inventory, consumer spending; drawbacks: intentions may not materialise; not forecasts per se (b) **Barometer / indicator approach** (c) **Economic model building** (GNP model / sectoral analysis)*

- → ***Indicators**: **leading** (turn before the economy) · **roughly coincidental** (turn with it) · **lagging** (turn after it). Show direction, not magnitude; may conflict → **diffusion / composite index** (hard to compute; doesn't remove irregular movements). **Money supply** ↑ → inflation; stocks a hedge → prices rise*
- → ***GNP model steps**: (i) hypothesise total demand (GNP) from political stability, inflation, economic levels (ii) forecast components — consumption, gross private domestic investment, government purchases, net exports (iii) add them (iv) compare with an independent agency's forecast and test consistency*

THEORY

③ Industry analysis (SM 1.3)

Industries don't grow at the same rate → assess demand, cost structure, government constraints.

Factor (7)	SM
(a) Product life cycle	High profit in initial & growth stages, medium but steady in maturity, sharp decline at the end
(b) Demand-supply gap	Excess supply cuts profit (lower price); shortage raises it
(c) Barriers to entry	Innate to product / technology, or created by existing firms
(d) Government attitude	Crucial determinant of prospects
(e) State of competition	Leaders, homogeneous vs differentiated products, demand by customer / area, growth / cyclical / defensive / decline industry
(f) Cost conditions & profitability	Cost allocation & controllability, price, capacity (installed / idle / operating), capex needed; ratios — GP, OPM, ROE, return on capital
(g) Technology & research	Nature of technology, expected changes, capex-to-sales over time, R&D spend, sales & profit in short / immediate / long run

Techniques: (a) **regression & correlation** on demand drivers (GNP, disposable income, per-capita consumption, price elasticity) (b) **input-output analysis** — flow of goods through the economy to spot growing / declining industries.

④ Company analysis (SM 1.4) — 13 points

- **(a) Net worth & book value** — equity + preference + free reserves − intangibles − losses; BV may not reflect worth: price reflects **future earnings**; BV on **historical cost**
- **(b) Sources & uses (funds flow)** — short-term funds financing long-term assets → liquidity crunch
- **(c) Cross-sectional & time series** — common-size statements, ratio analysis
- **(d) Size & ranking** — capital employed, profit, ROI, sales vs peers; know-how, R&D, price leadership
- **(e) Growth record — P/E**, % growth in EPS, % growth in **net block** (capacity). SM: A ₹150 / EPS 25 = P/E 6 vs B ₹250 / 25 = 10 → A preferable if all else equal. "**Growth is the single most important factor**"
- **(f) Financial analysis** — key: EPS, P/E, yield, book value, intrinsic value (10 years, vs industry); solvency (current, liquid, D/E)
- **(g) Competitive advantage** (Apple's brand, Samsung in mobiles) · **(h) Quality of management** — experience, skills, integrity; earnings calls, MD&A, insider selling, 5-year record · **(i) Corporate governance** — LODR 2015 compliance, stakeholder policies, disclosure quality & timeliness, independent directors · **(j) Regulation** — Companies Act, SCRA, SEBI Act; FEMA / RBI · **(k) Location & labour relations** · **(l) Pattern of shareholding** (PNB nationalisation — institutions raised compensation) · **(m) Marketability** — public holding, speculative interest, exchange, volume

Techniques: correlation & regression (simple / multiple) · trend analysis · decision tree (multiply probabilities along each sequence, then add). Also **group analysis** — SEBI disclosure of group companies; RBI group exposure norms.

TYK THEORY Q2 & Q3

5 Technical analysis (SM 2.1–2.3)

- → Studies price charts assuming trends **repeat** (investor psychology follows patterns); studies investor and price behaviour. Asks: (i) is there a trend? (ii) will it reverse? Methods: **visual** (charts) and **statistical**
- → **Assumptions (4)**: (i) value depends on **supply & demand** (ii) S & D governed by rational and irrational factors (iii) prices move in **trends** lasting a substantial time (iv) relies on charts, not financial statements
- → **Principles (3)**: (a) **market discounts everything** (b) **price moves in trends** (c) **history repeats itself**

6 Dow theory (SM 2.4.1) — TYK Q2

Charles Dow (founder of Dow Jones), late 19th century; measures relative strength of the market; a business barometer. Based on **DJIA** and **DJTA**. Three simultaneous movements: **primary** (main bull / bear trend, 1 year to 36 months or longer) · **secondary** (opposite direction, 2 weeks to a month or more) · **daily** fluctuations (not interpreted, but studied). Tells **where the market is going, not how far**. Successively higher highs & higher lows → bull; lower highs & lows → bear.

Three moves up: (1) accumulation by far-sighted investors (2) first reports of good earnings (3) widespread reports of well-being + rampant speculation. Then far-sighted investors sell → first move down; non-sustainability confirmed → second; **distress selling** → third.

7 Elliot wave theory (SM 2.4.2) — TYK Q3

Ralph Elliot, 1934, inspired by Dow theory and nature; 75 years of price data. Wave = price movement from one change in direction to the next in the same direction, from buying / selling impulses. **Impulsive (basic) waves**: 3 or 5 waves in the main direction (bull or bear). **Corrective (reaction) waves**: 3 waves against it. Waves 1, 3, 5 directional; 2 & 4 corrective. **Complete cycle = 5-wave impulse + 3-wave correction (a, b, c) = 8 waves**, bullish and bearish phases; then a fresh cycle.

8 Random walk theory (SM 2.4.3)

Prices are statistical ups and downs, **unpredictable**; no link between present and future prices; successive changes independent. **M. G. Kendall** (SM: "Kendell") — like a roulette wheel; "drunk walking in a blind lane". (a) prices can't be predicted (b) trends are a statistical expression of past data (c) no connection between successive peaks and troughs.

9 Charts (SM 2.5)

Line (closing prices joined) · **Bar** (vertical high-low bar; ticks for open & close) · **Japanese candlestick** (white = close > open = bullish; black = close < open = bearish; **Doji** = open ≈ close) · **Point & figure** (detects reversals; choose **box size** — smaller = more sensitive — and **reversal criterion**; X rising, O falling).

THEORY + NUMERICAL

10 Market indicators (SM 2.6)

- → **Breadth index** = net advances or declines ÷ issues traded; supports Dow averages = technical strength; contradicts = weakness (reversal)
- → **Volume**: rising price + rising volume → buy (unsatisfied demand); falling + rising volume → bear. SM also: rising market with decreasing volume indicates a bull market; falling with dwindling volume a bear market. Use with Dow theory
- → **Confidence index** = high-grade bond yields ÷ low-grade bond yields; rising precedes a rising market; usually (not always) a **leading** indicator
- → **Relative strength**: some securities rise faster in bull / fall slower in bear and keep that strength — measured by historical returns or ratio to industry / market
- → **Odd-lot theory**: **contrary-opinion** — average person usually wrong; predicts tops in bull markets and reversals

11 Support, resistance & patterns (SM 2.7–2.8)

Peak after a fall = **resistance** (selling pressure); trough after a rebound = **support** (buying pressure); breach = new levels. SM eg: A ₹100 / ₹125; B ₹140 / ₹160.

Pattern	Signal
Channel	Uniform tops & bottoms; up-sloping = rising, down-sloping = falling
Wedge	Tops and bottoms move in opposite directions, or same direction at different rates
Head & shoulders (top)	Most important **reversal** pattern; **bearish** — below neckline → fall ≈ head-to-neckline distance → **sell**
Inverse head & shoulders	**Bullish** — above neckline → **buy**
Triangle / coil	Uncertainty; breakout direction hard to predict
Flags & pennants	Previous trend likely to **continue**
Double top / double bottom	Bearish / bullish
Gap	Open vs previous close; wider gap = stronger continuation signal

12 Moving averages (SM 2.9)

$AMA_{n,t}$ = $(P_t + \dots + P_{t-(n-1)}) \div n$. Trend: **200-day** (or 30-week) = long term · **60-day** = intermediate · **10-day** = short term

$EMA_t = a \times P_t + (1 - a) \times EMA_{t-1}$, $a = 2 \div (n + 1)$ — latest price weighted most

Buy signal	Sell signal
(a) price rises through a flattening MA (b) price falls below a **rising** MA (c) price above MA falls but turns up before reaching it	(a) price falls through a flattening MA (b) price rises above a **falling** MA (c) price below MA rises but turns down before reaching it

ICAI paper (RSI): RSI = 100 − 100 ÷ (1 + RS), RS = avg gain ÷ avg loss; update averages as (previous × 13 + today) ÷ 14. >70 overbought · 50–70 bullish · 30–50 bearish · <30 oversold (ICAI answer: 34.64 → bearish).

TYK THEORY Q1

13 Evaluation of technical analysis (SM 2.10) & FA vs TA (SM 3)

For (3)

(a) crowd psychology makes trends persist; spotted early (b) demand-supply shifts are gradual (c) information is assimilated over time, so price keeps moving one way

Against (4)

(a) no convincing explanation of tools (b) evidence for random walk (c) by the time a trend is signalled it has happened (d) **self-defeating** — more users, less value

Efficient market → TA may not work; real markets are imperfect → useful **with** fundamental analysis. **FA vs TA**: method — macro (GDP, inflation, rates) + micro (sales, profitability, solvency) vs historical market data (price, volume, open interest) · rule — "prices discount everything" vs "price captures everything" · use — **long-term** vs **short-term** investing.

14 Efficient market theory (SM 4) — TYK Q1

- **Eugene Fama** (Chicago), 1960s: all price-sensitive information is fully reflected in prices → no one can consistently beat the market; higher return only by taking **more risk**
- **Why random**: information free & instant to all · keen competition makes prices reflect intrinsic values · prices change only on **new, unpredictable** information
- **Misconceptions (SM 4.2)**: (1) EMH doesn't mean prices always equal fair value — the future is uncertain, surprises keep prices fluctuating (2) managers' failure to outperform doesn't mean incompetence — markets are efficient **because** managers compete well (3) random movement ≠ irrational market — **rational, competitive investors make price changes random**
- **Levels**: **weak** — past prices & volumes · **semi-strong** — + all public information · **strong** — all information, public and private
- **Weak-form tests**: **serial correlation** (none found, short-term) · **run test** (runs vs a random series — supports random walk) · **filter rule** (buy after N% rise, sell after N% fall — doesn't beat buy-and-hold after commissions)
- **Semi-strong**: Fama, Fisher, Jensen & Roll — 940 NYSE splits 1957–59; Ball & Brown (SM: "Boll") — earnings announcements. **Anomalies**: slow adjustment to quarterly earnings surprises · small firms beat large · low P/E beats high P/E · **Monday** returns lower
- **Strong**: insiders and exchange specialists earn superior returns; **mutual fund managers do not** (Burton Malkiel, *A Random Walk Down Wall Street*)
- **Challenges (4)**: (a) information inadequacy / misinformation (b) limited processing capacity (Herbert Simon; David Dreman) (c) irrational behaviour (Keynes; **L. C. Gupta** — "blind man firing a gun") (d) monopolistic influence of big institutions

Equity research (SM 5): buy / sell / hold calls, also used in M&A swap ratios. Tools: Bloomberg Terminal (1981), Benzinga Pro, Refinitiv Eikon, MarketXLS, Stockopedia, Koyfin, Finbox.io, GuruFocus, Business Quant, YCharts.

15 Correlation — how Ch 4 actually connects

Ch 4 idea	The actual link (app-written cross-reference; the Ch 4 side is SM)
Intrinsic value, Gordon	**Ch 5**: same DDM; P/E = payout ÷ (k − g). **Ch 14**: equity research feeds the swap ratio.
EMH levels	**Ch 6 / Ch 8**: efficient markets favour **passive** (index / ETF) strategies; MF managers not beating the market = strong-form evidence; Jensen's α measures any "beating".
Leading indicators, money supply, inflation	**Ch 2**: interest-rate & currency risk identifiers; **Ch 3**: government-policy impact on capital budgeting.
Odd-lot, confidence index	**Ch 5**: confidence index uses bond yields — high-grade vs low-grade = credit spread.